Run report: GAG presentation matched study, run-001

Study `research/gag_presentation_matched_20261031` (no ledger EXP id; protocol `GAG-PRESENTATION-MATCHED-20261031` v1, frozen sha256 `609c701c…5ad2` at commit `7635a467`, filed before any study data). **Lead** `LEAD-ICE-gag-contraction-planner`. **Run** `runs/run-001`, 2026-11-01 17:55:01–17:55:57 UTC, code commit `bfccbce5`, clean tree. **Export** `runs/run-001/export.json` (`check-export --verify-files`: 0 findings).

What ran

Gaudry/Diem decomposition systems at $n = m = 3$ over $\mathbb{F}_{p^3}$, factor base = points with abscissa in $\mathbb{F}_p$, built by `descent.descent_system` and solved by msolve 0.6.5-1build2 (sha256 `c2722288…`) with `-c 0 -P 1 -t 1 -v 2`. Primes 1009, 4099, 32749; ten planted instances per prime (`design.json` instance rule), one warmup instance (1009, 99) excluded. Four arms per instance, order shuffled per instance: **chain** (baseline, Semaev's chained system, 6 unknowns, degree 2), **tree** (merge tree, 6 unknowns), **monolithic** (wire contracted over $\mathbb{F}_q$ to S_4, then descended: 3 unknowns, degree 4 per variable), **chain-aa** (chain with a different separating-form stream: the A/A control).

Command: `PYTHONPATH=src python3 research/gag_presentation_matched_20261031/driver.py --protocol …/protocol.json --out …/runs/run-001`, then the frozen `analyze.py`.

Host: shared 2-vCPU cloud container (Intel Xeon 2.1 GHz, kernel 6.18.44), no isolation; load average 4.13 at start, 2.35 at end. The S3 resume-checkpoint gate failed (no upload token in the session; `checkpoint_gate.txt`); the owner waived it for this run before launch.

Flow

Planned 124, attempted 124, completed 124, not completed 0, interrupted 0, resumed 0, excluded 0. Every arm: 30 measured rows + 1 warmup. One execution needed two msolve attempts (`0035-p1009-i7-chain-aa`: first attempt reported a non-separating form); both attempts are charged, as the protocol requires.

Exact checks (claim C1): hold

All 30 instances: every arm returned status `ok`, ideal degree 384, and the same set of relations verified by point addition over $\mathbb{F}_{p^3}$; the planted decomposition was in it on every row. (Three instances at $p = 4099$ have two decompositions; all arms return both.)

Counted measurements (measured; medians over 10 instances, min–max in brackets)

arm	F4 cells, p=1009	p=4099	p=32749	F4 max degree / rounds
chain	11 795 457	11 795 457 [11 792 928 – 11 795 457]	11 795 457	8 / 20
tree	12 911 317 [12 909 180 – 12 911 317]	12 911 317 [12 848 210 – 12 911 317]	12 911 317	8 / 20
monolithic	1 298 729 [1 298 254 – 1 298 729]	1 298 729 [1 298 541 – 1 299 222]	1 298 729	15 / 33
chain-aa	11 795 457 [11 790 746 – 23 590 914]	11 795 457 [11 792 928 – 11 795 457]	11 795 457	8 / 20

F4 cells = sum over msolve's printed F4 rounds of rows × columns, summed over attempts. The chain-aa maximum 23 590 914 at $p = 1009$ is the two-attempt execution.

Comparisons (derived by the frozen analyze.py)

Per prime: median over instances of $\log_2$(arm cells / chain cells); percentile bootstrap interval at level $1 - 0.05/6$ (20 000 resamples); exact paired sign-flip test.

comparison	median ratio (each prime)	bootstrap interval, log2	sign-flip p	A/A max |log2|
monolithic / chain	0.1101, 0.1101, 0.1101	[−3.183, −3.183] at all three	0.00195 (each)	1.000, 0.000, 0.000
tree / chain	1.0946, 1.0946, 1.0946	[0.130, 0.130] at all three	0.00195 (each)	1.000, 0.000, 0.000
chain-aa / chain	1.0000, 1.0000, 1.0000	[0, 0]	1.0	—

Verdicts (protocol vocabulary). C-mono: **supported_within_scope**: the monolithic (S_4) presentation used 0.110 times the chain's F4 matrix cells (about 9.1 times fewer) at every prime, beyond the A/A spread. C-tree:

inconclusive: the tree used 1.095 times the chain's cells at every prime, but at $p = 1009$ the A/A maximum $|\log_2|$ is 1.0, set by the two-attempt chain-aa execution, so the frozen rule does not pass at that prime; at the other two primes the A/A spread is 0.

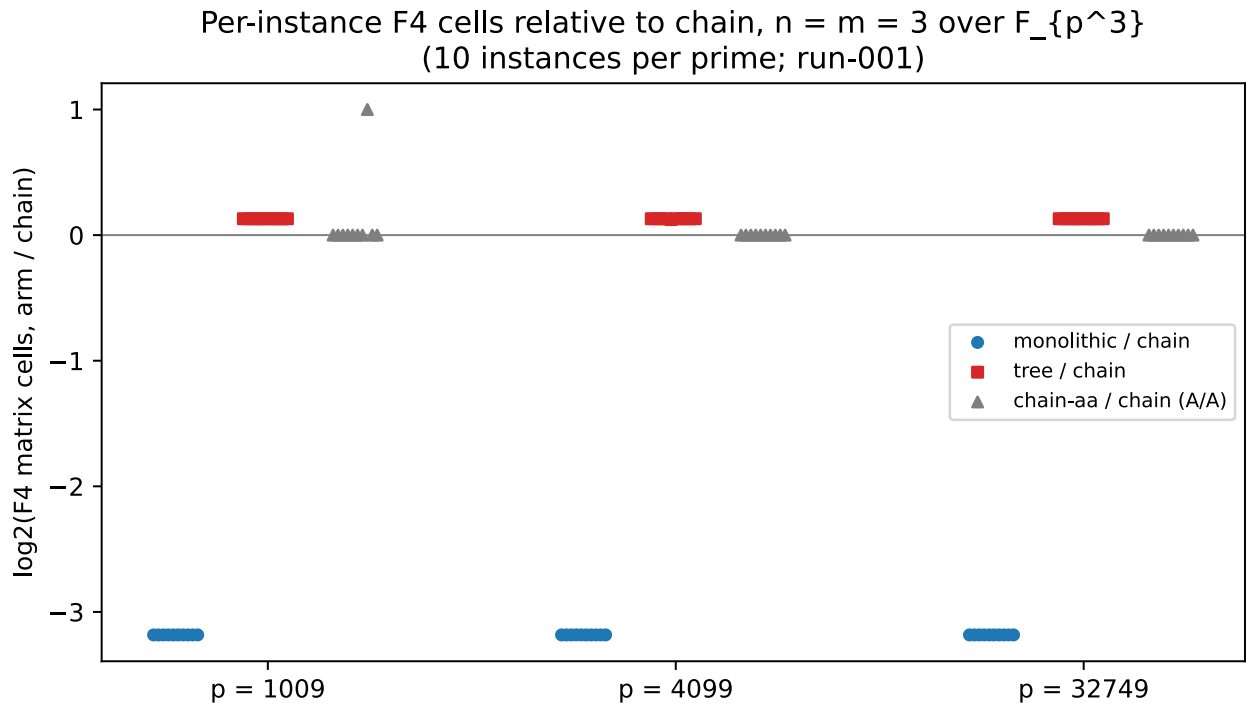


Figure 1: Per-instance $\log_2$(arm / chain) F4 matrix cells, 10 instances per prime, run-001 (`runs/run-001/analysis.json`). The A/A point at +1.0 ($p = 1009$) is the execution charged with two msolve attempts.

Secondary, descriptive only (wall time, isolation none)

Median msolve subprocess time per execution: chain 0.22–0.23 s, tree 0.24–0.25 s, monolithic 0.031–0.032 s, chain-aa 0.22–0.23 s. Not a result (protocol: wall time decides nothing on an unisolated host).

Scope

This describes msolve 0.6.5's F4 on these $n = m = 3$ toy instances over $\mathbb{F}_{p^3}$ for three primes below 2^{16}. It makes no ECDLP cost, scaling, other-solver or other-(n, m) statement, and charges no point extraction or Python system build. It is the opposite direction from the characteristic-2 Boolean report recorded in `KN-TECH-b18366` (chaining up to 50 times faster), in a different field, encoding and solver. $n = m = 4$ was not run (memory).

Next step

Evidence review of run-001 (`review-evidence`), then the $n = m = 4$ chain vs tree cell on a host with more than 8 GB, and the prime-field $m = 3, 4$ arms named in the lead.